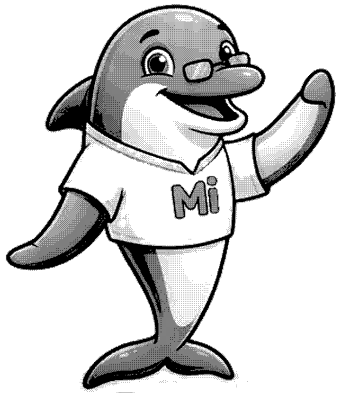
Math Inquirers

# Extension Workbook

## with answers

## Level 3

Self-study | Ages 8-10




My name _____

# About this book

**Math Inquirers Extension Workbook, Level 3**
Self-study practice with answers.


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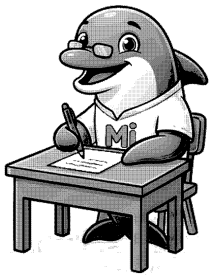
- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

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# Welcome!

This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

### How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

### For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 15: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- The rulers in Unit 9 are printed at real size.

### The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

# Contents


| | | |
|---|---|---|
| | **About this book**<br>Copyright and licence | page 2 |
| | **Welcome** | page 3 |
| | **My pace plan** | page 5 |
| 1 | **Number sleuths (numbers to 10 000)**<br>Not in this proof edition | – |
| 2 | **Cryptarithm crackers**<br>Not in this proof edition | – |
| 3 | **Times table treasure**<br>Not in this proof edition | – |
| 4 | **Model makers (bar models)**<br>Not in this proof edition | – |
| 5 | **Remainder riddles**<br>Not in this proof edition | – |
| 6 | **Fraction walls**<br>Not in this proof edition | – |
| 7 | **Area architects**<br>Not in this proof edition | – |
| 8 | **Angle and line detectives**<br>Angle and line detectives · follows Holiday Program Lesson 8 · answers on page 16 | page 6 |
| 9 | **Measure and money masters**<br>Not in this proof edition | – |
| 10 | **Puzzle day**<br>Not in this proof edition | – |
| | **My progress** | page 13 |
| | **My certificate** | page 14 |
| | **Answers**<br>Check your work, with short reasons | page 15 |

# My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

| Week | Unit | Session 1<br>Learn it, Start here | Session 2<br>Start here | Session 3<br>Think harder, Check yourself | Weekend page<br>Go further | Done |
|---|---|---|---|---|---|---|
| 1 | Number sleuths (numbers to 10 000) | Not in this proof edition | | | | ☐ |
| 2 | Cryptarithm crackers | Not in this proof edition | | | | ☐ |
| 3 | Times table treasure | Not in this proof edition | | | | ☐ |
| 4 | Model makers (bar models) | Not in this proof edition | | | | ☐ |
| 5 | Remainder riddles | Not in this proof edition | | | | ☐ |
| 6 | Fraction walls | Not in this proof edition | | | | ☐ |
| 7 | Area architects | Not in this proof edition | | | | ☐ |
| 8 | Angle and line detectives | 6-7 | 8-9 | 10 and 12 | 11 | ☐ |
| 9 | Measure and money masters | Not in this proof edition | | | | ☐ |
| 10 | Puzzle day | Not in this proof edition | | | | ☐ |
| 11 | Catch-up week | Repeat any Check yourself where your score was 2 or less. | | | | ☐ |

### Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

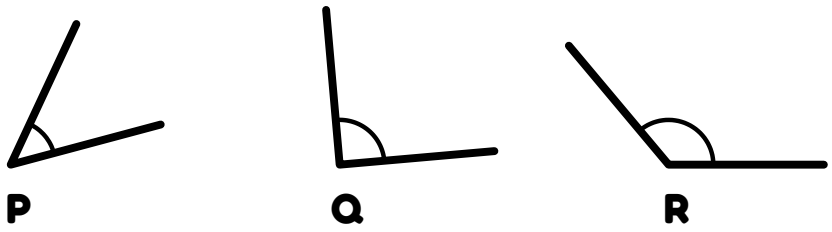
# Learn it: Angles, perpendicular lines and parallel lines

**My goal. I can compare angles with a right angle. I can find perpendicular and parallel lines. I can count angles in a figure.**

An angle is made where 2 lines meet. A right angle is a square corner, like the corner of a page. Use the corner of a piece of paper as a right angle checker.

## Worked example

**Use a right angle checker. Is each angle smaller than, equal to, or greater than a right angle?**




1. **Put the paper corner on the point where the lines meet.**
2. **Line up one side of the paper with one line.**
3. **Look at the other line. If it is inside the corner, the angle is smaller. If it is outside, the angle is greater.**

**Answer: P is smaller than a right angle. Q is a right angle. R is greater than a right angle.**

## Lines to remember

1. Perpendicular lines meet at a right angle.
2. Parallel lines never meet. They stay the same distance apart.
3. Count angles in order: the small ones first.

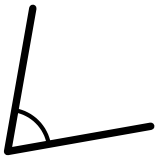
## Your turn

Answer at the back of the book.

**Try**

**Use a right angle checker. This angle is**

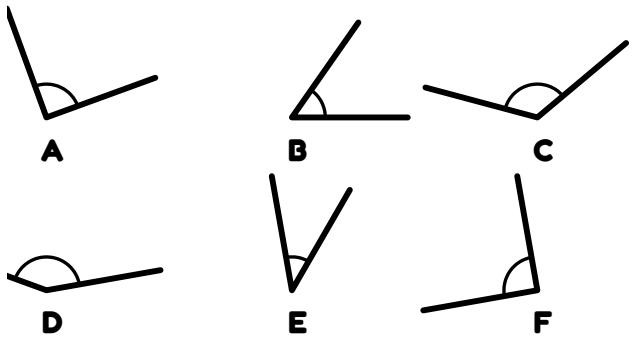
**smaller than** **equal to** **greater than** **a right angle.**



## Start here

Do these first. Take your time.

**1** **Use a right angle checker on each angle. Write the letters.**

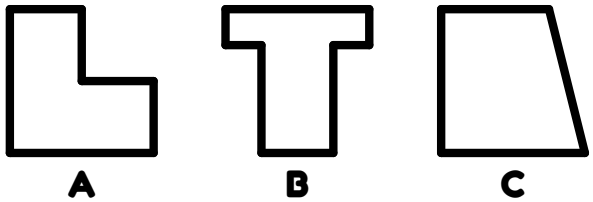



(a) Smaller than a right angle: _____

(b) A right angle: _____

(c) Greater than a right angle: _____

**2** **Count the right angles INSIDE each shape. Check each corner. A dent is greater than a right angle, so leave it out.**




(a) Shape A has ☐ right angles.

(b) Shape B has ☐ right angles.

(c) Shape C has ☐ right angles.

**3** **Look at the letters.**

F V N L

(a) Circle each letter with 2 PARALLEL lines. F V N L

(b) Circle each letter with 2 PERPENDICULAR lines. F V N L

## Start here

Keep going. Check your work.

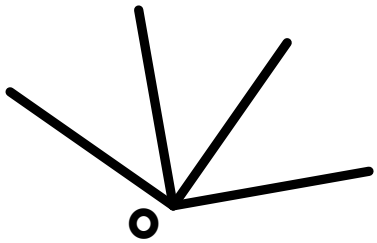
**4** **These 4 lines all start at point O. Count the angles. Count the small ones first.**

(a) Angles made of 1 small angle: ☐

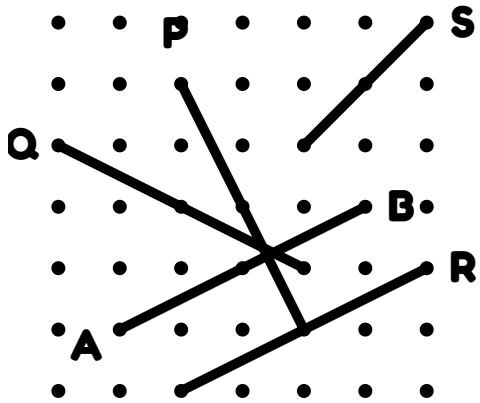
(b) Made of 2 small angles: ☐

(c) Made of 3 small angles: ☐

(d) Angles in all: ☐




**5** **Look at the lines on the dot grid.**



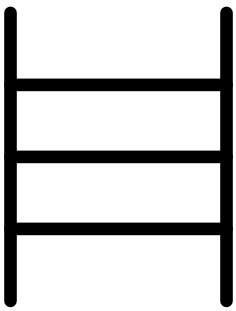

(a) Which line is PARALLEL to line AB? P Q R S

(b) Which line is PERPENDICULAR to line AB? P Q R S

**6** **A ladder has 2 upright rails and 3 rungs. Each rung meets a rail at right angles.**

(a) Where one rung meets the 2 rails there are ☐ right angles.

(b) The ladder has ☐ right angles in all.



## Start here

Nearly there. You can do it.

**7** **Mei faces north. She makes turns of a right angle.**

**(a)** She turns 1 right angle to the right. She faces N E S W

**(b)** From north, she turns 3 right angles to the right. She faces
N E S W

**(c)** From north, she turns 2 right angles to the left. She faces
N E S W

**8** **Fill in the blanks.**

**(a)** A rectangle has ☐ right angles.

**(b)** A rectangle has ☐ pairs of parallel sides.

**(c)** A triangle has at most ☐ right angle.

**(d)** Two sides that meet at a corner of a square are
parallel perpendicular

**9** **Circle True or False.**

**(a)** Lines that are perpendicular cross at a right angle. True False

**(b)** Parallel lines meet if you make them long enough. True False

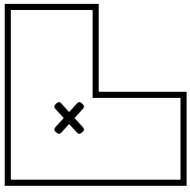
**(c)** One line can be perpendicular to 2 parallel lines. True False

**(d)** Two lines that cross are always perpendicular. True False

**10** **Siti says angle X is a right angle.**

**(a)** Is she right? Yes No

**(b)** Angle X is
smaller than equal to greater than
a right angle.



## Think harder

Hints are upside down. Try first!

**1** **Lines from a point O make angles. 2 lines give 1 angle, 3 lines give 3 angles and 4 lines give 6 angles.**

**(a)** With 5 lines there are ☐ angles.

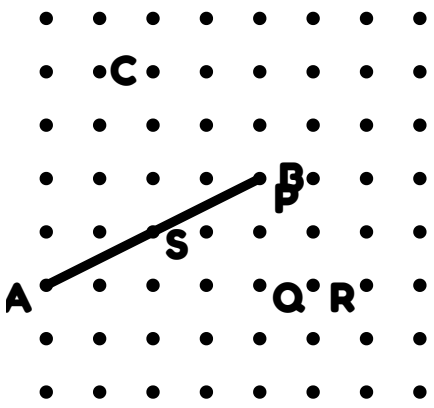
**(b)** With 7 lines there are ☐ angles.

5 lines: _____

7 lines: _____

**Hint.** Small angles first: 4. Then angles made of 2 small ones: 3.

**2** **Draw a line from C that is PERPENDICULAR to line AB. Use a ruler.**

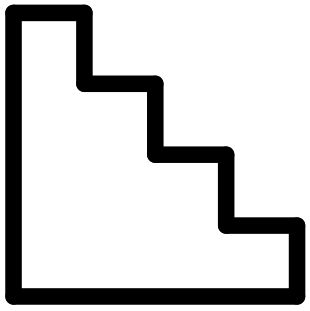



**(a)** Your line passes through dot

P Q R S

**Hint.** AB goes 2 across and 1 up. Turn it a quarter turn.

**3** **A staircase has 4 steps.**



**(a)** It has ☐ right angles inside.

**(b)** It has ☐ dents.

**(c)** A staircase with 8 steps has ☐ right angles inside.

**Hint.** Count the dents first. Each new step adds 1 right angle.

## ★ Go further

A weekend page. Big challenges. Hints are upside down.

**1** **Beyond Level 3: Degrees.**
**A right angle is 90 degrees.**

**(a)** 2 right angles make a straight line: ☐ degrees.

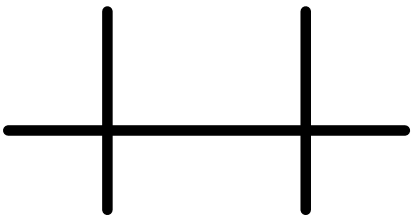
**(b)** 4 right angles make a full turn: ☐ degrees.

**(c)** Half of a right angle is ☐ degrees.

**(d)** 3 right angles make ☐ degrees.

**Hint.** Add the right angles: 90 + 90 + ...

**2** **Beyond Level 3: Lines and a third line.**
**Two upright lines both cross the long line at right angles.**



**(a)** The 2 upright lines are parallel perpendicular

**(b)** A line is perpendicular to one of 2 parallel lines. Is it also perpendicular to the other? Yes No

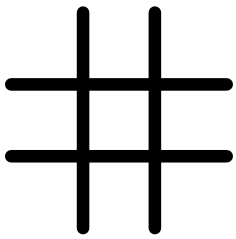
**Hint.** Do the 2 upright lines ever meet?

**3** **A board has 2 lines across and 2 lines up.**

**(a)** There are ☐ pairs of parallel lines.

**(b)** The lines make ☐ right angles where they cross.

**(c)** A board with 3 lines across and 3 lines up makes ☐ right angles.



**Hint.** Each place where 2 lines cross makes 4 right angles.

## ✔ Check yourself

No hints. Try each one on your own.

**1**

**Use a right angle checker. This angle is**

**smaller than** **equal to** **greater than** **a right angle.**



**2** **Circle each letter with 2 parallel lines.** **E** **H** **V** **T**

E H V T

**3** **A rectangle has** ☐ **pairs of parallel sides.**

Working: _____

**4** **Ben faces east. He turns 1 right angle to the left. He faces**

**N** **E** **S** **W**

Working: _____

**My score:** ☐ **out of 4. Check the answers at the back.**

### What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

### Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

# My progress

Tick each box when you finish. Write your Check yourself score.

| Unit | Topic | Learn it | Start here | Think harder | Go further | Score | Date |
|---|---|---|---|---|---|---|---|
| 1 | Number sleuths (numbers to 10 000) | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 2 | Cryptarithm crackers | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 3 | Times table treasure | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 4 | Model makers (bar models) | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 5 | Remainder riddles | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 6 | Fraction walls | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 7 | Area architects | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 8 | Angle and line detectives | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 9 | Measure and money masters | ☐ | ☐ | ☐ | ☐ | / 4 | |
| 10 | Puzzle day | ☐ | ☐ | ☐ | ☐ | / 4 | |

## Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____




# Certificate of Completion

## Well done,

**You finished the Math Inquirers Extension Workbook, Level 3.**

Date

Signed by a grown-up

# Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

# Unit 8 · Angle and line detectives

**Your turn**
page 6

**Answer:** smaller than
**Why:** The second line is inside the corner of the checker, so the angle is smaller than a right angle.

**START HERE**

**1**
page 7

**Answer: (a)** B and E **(b)** A and F **(c)** C and D
**Why:** A and F fit the checker exactly, so they are right angles. B and E are inside the corner of the checker, so they are smaller. C and D stick out past the corner of the checker, so they are greater.

**2**
page 7

**Answer: (a)** 5 **(b)** 6 **(c)** 2
**Why:** Shape A is an L. It has 5 right angles, and its dent is greater than a right angle. Shape B has 6 right angles and 2 dents. Shape C has right angles only at its 2 left corners. Its 2 sloping corners do not fit the checker.

**3**
page 7

**Answer: (a)** F and N **(b)** F and L
**Why:** F has 2 parallel arms and a stem that meets them at right angles. V has 2 sloping lines that are neither parallel nor perpendicular. N has 2 parallel upright lines, but its sloping line is not perpendicular to them. L has 2 lines that meet at a right angle.

**4**
page 8

**Answer: (a)** 3 **(b)** 2 **(c)** 1 **(d)** 6
**Why:** There are 3 small angles. The next size is made of 2 small angles side by side: 2 of those. Then 1 angle made of all 3 small ones. 3 + 2 + 1 = 6 angles.

**5**
page 8

**Answer: (a)** R **(b)** P
**Why:** Line AB goes 4 across and 2 up. R goes 4 across and 2 up too, so it never meets AB: parallel. P goes 2 across and 4 down, which turns AB through a right angle: perpendicular. Q and S slope in other ways.

**6**
page 8

**Answer: (a)** 4 **(b)** 12
**Why:** At each end of a rung, the rung and the rail make 2 right angles (one above the rung and one below it). A rung has 2 ends, so 4 right angles. 3 rungs make 3 × 4 = 12 right angles.

**7**
page 9

**Answer: (a)** E **(b)** W **(c)** S
**Why:** Going round to the right, north is followed by east, south and west. (a) 1 turn: east. (b) 3 turns: west. (c) 2 turns to the left from north: west, then south. Two right angles in either direction face the opposite way: south.

**8**
page 9

**Answer: (a)** 4 **(b)** 2 **(c)** 1 **(d)** perpendicular
**Why:** (a) Each corner of a rectangle is a right angle: 4. (b) The 2 long sides are parallel and the 2 short sides are parallel: 2 pairs. (c) If a triangle had 2 right angles, those sides would be parallel and never meet to make a triangle. (d) The sides of a square meet at right angles.

**9** page 9
**Answer: (a) True (b) False (c) True (d) False**
**Why:** (a) That is what perpendicular means. (b) Parallel lines never meet. (c) A line across 2 rails of a ladder is perpendicular to both. (d) Lines can cross at any angle. Only some crossings make right angles.

**10** page 9
**Answer: (a) No (b) greater than**
**Why:** Angle X is a dent. It points into the shape. A checker does not fit inside it: the angle is greater than a right angle.

## THINK HARDER

**1** page 10
**Answer: (a) 10 (b) 21**
**Why:** With 5 lines there are 4 small angles. Then 3 angles made of 2 small angles, 2 made of 3, and 1 made of 4. 4 + 3 + 2 + 1 = 10. With 7 lines the small angles are 6: 6 + 5 + 4 + 3 + 2 + 1 = 21.

**2** page 10
**Answer: (a) Q**
**Why:** AB goes 2 across for every 1 up. A line perpendicular to it goes 1 across and 2 down (a quarter turn). From C: 1 across and 2 down reaches the dot at (3, 4), then 1 across and 2 down again reaches Q.

**3** page 10
**Answer: (a) 7 (b) 3 (c) 11**
**Why:** Count the corners that fit a checker: 7. The 3 corners that dent into the shape are greater than a right angle. A staircase with n steps has n - 1 dents and n + 3 right angles. For 8 steps: 8 + 3 = 11 right angles.

## GO FURTHER

**1** page 11
**Answer: (a) 180 (b) 360 (c) 45 (d) 270**
**Why:** (a) 90 + 90 = 180. (b) 4 × 90 = 360. (c) 90 ÷ 2 = 45. (d) 3 × 90 = 270.

**2** page 11
**Answer: (a) parallel (b) Yes**
**Why:** (a) Two lines that are perpendicular to the same line stay the same distance apart and never meet: they are parallel. (b) A line that crosses 2 parallel lines makes the same angle with both. So a right angle with one means a right angle with the other.

**3** page 11
**Answer: (a) 2 (b) 16 (c) 36**
**Why:** (a) The 2 lines across are parallel and the 2 lines up are parallel: 2 pairs. (b) The lines cross in 2 × 2 = 4 places, and each crossing has 4 right angles: 16. (c) 3 × 3 = 9 crossings, 9 × 4 = 36 right angles.

## CHECK YOURSELF

**1** page 12
**Answer: greater than**
**Why:** The second line sticks out past the corner of the checker, so the angle is greater than a right angle.

**2** page 12
**Answer: E and H**
**Why:** E has 3 parallel arms. H has 2 parallel upright lines. V has 2 sloping lines that are not parallel. T has no pair of parallel lines.

**3** page 12

**Answer:** 2
**Why:** The 2 long sides are parallel and the 2 short sides are parallel.

**4** page 12

**Answer:** N
**Why:** Turning left from east, the next direction is north.